

you are a nutrition analyst

Below is a rigorous, evidence-based readout of your 1–15 Mar 2026 meal/exercise dataset, interpreted together with the CGM report and your body-composition snapshot. I'll call out *why* peaks happened (or didn't), what's most correlated with the size and timing of peaks, and what to try next.

1) What's in the data (and quality notes)

Coverage. I see **26 meal logs** and **exercise logs** across **9 days**. Of the meals, **20 have identifiable post-meal peaks** (peak value and time recorded). Mean post-prandial excursion (peak – pre-meal) was **~2.93 mmol/L**, median **2.9 mmol/L**; average **time-to-peak ~89 minutes**.

CGM context (15 days). Average glucose **5.9 mmol/L**, CV **16.9%**, eHbA1c **5.3%**, TIR (3.9–7.8) **93.2%**. Typical post-meal peaks: **Lunch 8.4** and **Dinner 8.0 mmol/L**; the report's "goal" for 2-hour excursion is **<2.2 mmol/L**. .pdf)

Body composition (Fitdays snapshot). Height **170 cm**, weight **68.1kg**, BMI **~23.6**, body fat rate **~20.7%**, skeletal muscle mass shown near **30.5 kg**, body score **75/100**, basal metabolism **~1540kcal**. This profile generally aligns with good insulin sensitivity for many people, which is consistent with your high TIR.

Data caveat. Several meals have no recorded "peak" despite visible bumps on the CGM daily charts (e.g., 14-Mar lunch shows ~6.5 mmol/L but no explicit "peak time" entry in the table). I treated "no peak" rows as *no detected peak value/time in the meal log*, not as proof that glucose was perfectly flat. .pdf)

2) Why peaks existed (and why they didn't)

A. The carbohydrate quality signal is the strongest

I tagged meals heuristically by likely glycemic impact based on the description (e.g., **rice/rice balls/fried rice, roti, fries, dumplings** → higher glycemic load; protein/veg/nuts/eggs → lower). Results:

	≡ Diet category (n)	≡ Mean excursion (mmol/L)
1	Higher-carb/refined (6)	3.77
2	Moderate/mixed (6)	2.98
3	Lower-carb/protein-forward (8)	2.28

Examples of largest spikes:

- **Beef + egg roti** (home-made): **+6.6 (5.7 → 12.3)** — 06-Mar dinner.

- **Pork with rice** (restaurant): +5.1 (6.1 → 11.2) — 09-Mar lunch.
- **Lamb kebab + fries** (restaurant): +4.1 (6.2 → 10.3) — 12-Mar lunch.
- **7-Eleven chicken breast ×2 + rice ball** (convenience): +3.7 (5.6 → 9.3) — 02-Mar lunch.
- **Fried rice** (restaurant): +3.4 (6.2 → 9.6) — 10-Mar dinner.

This pattern is exactly what controlled trials and meta-analyses show: **glycemic index/load** and **amount of digestible carbohydrate** dominate post-prandial peaks, and **more soluble fiber / fewer rapidly digested starches** blunt excursions.

Green bananas: On 07-Mar dinner you had **chicken breast + 2 green bananas**, excursion was modest ($\approx +2.0$ to 7.2). **Resistant starch (RS)** in unripe bananas is digested slowly and lowers glycemic response; trials with **native banana starch (RS2)** show lower fasting and post-challenge glucose peaks.

B. Where you ate also mattered (proxy for carb type/preparation)

Average excursions by "nature of food":

- **Restaurant** (n=6): +3.63 mmol/L
- **Home-made** (n=13): +2.55 mmol/L
- **Convenience store** (n=1): +3.7 (single row, not generalizable)

Restaurants often use sauces/sugars, refined starches, and frying—consistent with higher peaks.

C. Timing of peaks: mostly ~70–105 min

Your **median time-to-peak** ~85 min (IQR ~69–105). The earliest notable peak (45 min) was the **rice ball lunch**, and the longest (240 min) was a **mixed home-cooked dinner** that likely had extended eating/late snacking. Earlier/larger peaks are typical with higher-GI foods; slower, smaller peaks come with protein/fiber/RS.

D. Why peaks sometimes didn't occur

Meals with **no logged peak value** include:

- 01-Mar dinner (**Mixed**, restaurant),
- 05-Mar lunch (**duck/chicken/egg**, home),
- 05-Mar dinner (**carrot, beef, dumpling**, home),
- 06-Mar lunch (**chicken, egg, nuts**, home),
- 10-Mar lunch (**7-Eleven chicken breast ×3**, convenience),
- 14-Mar lunch (**hotpot + dessert**, restaurant).

Likely reasons (supported by your CGM charts and dietary science):

1. **Low glycemic load meals** (e.g., chicken/eggs/nuts or chicken breast alone) → rises too small to be flagged as peaks. .pdf)
2. **Resistant starch / fiber effects** (green bananas earlier that day; vegetables) → delayed/flattened response.
3. **Grazing / prolonged meals** (hotpot): glucose might undulate below your personal "peak" threshold; the daily page shows lunch ~6.5 on 14-Mar without a formal peak record. .pdf)
4. **Measurement/logging gaps** (no peak time recorded) despite visible bumps (e.g., 10-Mar lunch chart shows ~7.4). .pdf)

3) Exercise timing and its effect

From your logs, **cardio/strength sessions** were typically **>60 min away from meals**; only a few fell within a strong glucose-modulating window. In this dataset we saw **no meals with cardio in the first 60 minutes** post-meal; one had pre-meal exercise within 2 hours. Consequently, the simple within-dataset comparison shows **no detectable mean reduction** from “cardio in 0–60 min” (since there were 0 such pairings here). Where there was **pre-meal exercise ≤ 2 h**, the mean excursion looked **slightly lower** (~ 2.9) than meals without (but $n=1 \rightarrow$ *not conclusive*).

What high-quality evidence says (and how you can use it):

- **10–15 minutes of light walking immediately after a meal** can significantly **lower the peak** compared with resting or walking later. Randomized crossover trials (young adults and people at risk) consistently show **smaller peaks/AUC** when walking starts **right after** eating.
- In **type 2 diabetes**, advice to walk **10 min after each main meal** beats generic “30 min/day” for reducing **post-meal** glucose.
- **Regular training**—both **aerobic** and **resistance**—improves **post-prandial glycemia** and **insulin sensitivity**, even independent of weight loss; resistance training adds muscle “glucose sink” capacity.
- Meta-analyses suggest combined **resistance + aerobic** routines may be optimal; cycling and resistance rank well for insulin sensitivity endpoints.

Practical read-through for your logs: on **07-Mar lunch** you ran **~ 66 minutes** after eating (slightly outside the strongest 0–60 min window) and peaked **8.1**; a similar lunch preceded by or followed immediately by **10–15 min walk** would likely shave the peak further, based on trials. .pdf)

4) Other correlates inside your data

- **Pre-meal level vs excursion:** essentially **no correlation** ($r \approx -0.01$). In your case, *what you eat* mattered far more than starting level.
- **Time-to-peak vs excursion:** also **no strong trend** ($r \approx -0.02$). High-GI items sometimes peaked earlier (e.g., rice ball at 45 min), but across all meals the relationship was weak.
- **Lunch vs dinner:** Lunch excursions averaged ~ 3.11 vs ~ 2.76 for dinner—consistent with your CGM “Daily meal-time glucose” page showing larger lunch peaks in this period. .pdf)

5) Targeted, actionable changes (ranked by expected impact)

Personal target (from CGM report): keep **2-hour excursions < 2.2 mmol/L** when possible. You are often close; a few meals (rice/roti/fries) exceeded that by a wide margin. .pdf)

1) Carb quality & portioning

- Swap/refine the “top spike” items:
 - **Roti** $\rightarrow$ **whole-grain/RS-enriched wraps**;
 - **Fried rice** $\rightarrow$ **steamed rice** (smaller portion) + **extra veg/protein**;
 - **Fries** $\rightarrow$ **salad/beans**;
 - **Rice ball** $\rightarrow$ **half portion** + **chicken/veg**.
- Keep leveraging **green (unripe) bananas** or **resistant-starch foods** (cooled rice/potatoes, legumes) when you want

fruit/starch with lower spikes.

2) Order & structure of eating

- **Protein/fiber first**, starch later; add **vinegar/veg** at the start. These tactics lower effective glycemic load and slow gastric emptying (supported across GI/GL literature).

3) Post-meal movement timing

- Make a 10–15 min easy walk begin within 0–10 min of finishing *higher-carb* meals. On busier days, even 3 × 5 min “movement snacks” help if you can’t do a continuous walk.

4) Strength training consistency

- Keep the 2–3×/week pattern (you logged multiple strength days). Over weeks, more muscle mass improves glucose uptake and reduces spikes from the same meal.

5) Restaurant playbook

- Choose **grilled/roasted protein + extra veg**, ask for **sauces on the side**, and swap **fries/rice** for **veg/beans**; split rice/dessert. Your data show **restaurant meals averaged +3.63 vs +2.55** at home.
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6) Quick “what happened” notes for specific days

- **06-Mar dinner (beef + egg roti)** → largest spike (12.3, +6.6). The **roti** (refined wheat) plus **fat/protein** likely delayed gastric emptying yet allowed a high total **glycemic load**, producing a substantial peak ~2 h later. Consider **half roti + more veg** or switch to a **high-RS wrap**; add **10-min walk** immediately after.
 - **09-Mar lunch (pork with rice)** → 11.2, +5.1. Classic high-GL starch serving. Try **half rice + extra non-starchy veg**, or replace rice with **beans**; include the “walk now” habit.
 - **07-Mar dinner (chicken + green bananas)** → 7.2, +2.0. Likely benefit from **resistant starch**; keep using this tactic when you want fruit with protein.
 - **05-Mar (both meals) & 10-Mar lunch (chicken breast ×3)** → **no logged peaks**; lower-GL pattern (protein-forward, minimal starch) explains the flat curves that day.
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7) How the CGM summary supports the story

- Your **average** (5.9) and **TIR 93%** indicate very good overall control through the fortnight; the **meal-time page** shows **lunch peaks highest** (8.4), matching the dataset where lunch excursions were larger than dinner. The report’s **2-hour target** (<2.2) gives a clear threshold to aim for when testing tweaks. .pdf)
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8) What I recommend tracking next (to tighten the analysis)

- 1) **Exact carb grams** (estimate is fine) and **added sugars** per meal.
- 2) **Exercise start time relative to meal** (e.g., T+8 min walk, 12 min, ~50–60% VO₂max pace).
- 3) **Meal duration** (continuous eating vs single sitting).
- 4) **Alcohol and sleep quality** (both can modulate post-prandial response).

We can then run a proper **multivariate model** to quantify the contribution of starch amount, GI proxy words, meal duration,

and exercise timing to excursion amplitude and time-to-peak. I can also generate visual dashboards (by weekday, restaurant vs home, etc.) on request.

9) TL;DR — your personalized “peak control” checklist

- **Trim/refactor refined starches** (rice/roti/fries/dumplings). Replace part with veg/beans; if keeping them, **front-load protein/fiber** and walk 10–15 min immediately after.
 - **Leverage resistant starch**: green bananas, cooled starches, legumes.
 - **Keep strength work** (2–3×/wk) and add **micro-walks** after meals on restaurant days.
 - Use your CGM’s **<2.2 mmol/L@2h** as the live feedback target when experimenting. .pdf)
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If you’d like, I can (1) plot your **excursion vs. food type** and **exercise timing** as charts, (2) add a **restaurant decision guide** tailored to your usual spots, and (3) set up a **one-page weekly scorecard** that flags any meals projected to exceed **+2.2**.